

Stable Working-Memory RNN Model Summary

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Purpose

This report summarizes the two current working-memory model baselines before adding noise or psilocybin-inspired perturbations. The baseline-delay model is the canonical categorical delayed-response model and uses a bounded tanh recurrent activation. The tuned-delay-stable model is the newer continuous circular-location delayed-response model with population tuning curves and ring-attractor analyses.

The results should be read as model-analysis evidence, not biological claims. The tuned model supports a sampled ring-like attractor interpretation, but it does not exhaustively map the full 64-dimensional hidden-state space.

1 Task and Model Overview

Property	Baseline delay	Tuned delay stable
Task type	Categorical cue-delay-response memory task	Continuous circular-location cue-delay-response memory task
Input representation	One-hot class cue plus context channel	Population bump over 32 angle-tuned units plus context channel
Output target	Correct class label	Reconstructed population bump decoded into an angle
Training objective	Cross-entropy, scored during response	Population MSE, with delay-period scoring enabled
Recurrent activation	Tanh recurrent activation	Tanh recurrent activation
Main performance metric	Accuracy	Circular angular error in degrees

Table 1: High-level comparison of the current categorical baseline and the tuned continuous baseline.

2 Baseline Delay

The baseline-delay model is the current categorical baseline. It uses tanh recurrent dynamics and solves the categorical task perfectly across the tested delay sweep. Its hidden-state speed drops substantially during the delay, indicating that the state slows and settles rather than continuing to accelerate.

Metric	Value
Held-out categorical accuracy	1.000
Accuracy at delay 20	1.000
Accuracy at delay 40	1.000
Accuracy at delay 60	1.000
Accuracy at delay 80	1.000
Early delay speed	0.561
Late delay speed	0.0349
Delay settling ratio	0.0623

Table 2: Baseline-delay headline metrics. The delay settling ratio is late-delay speed divided by early-delay speed. Values well below 1 indicate slowing and settling during the delay.

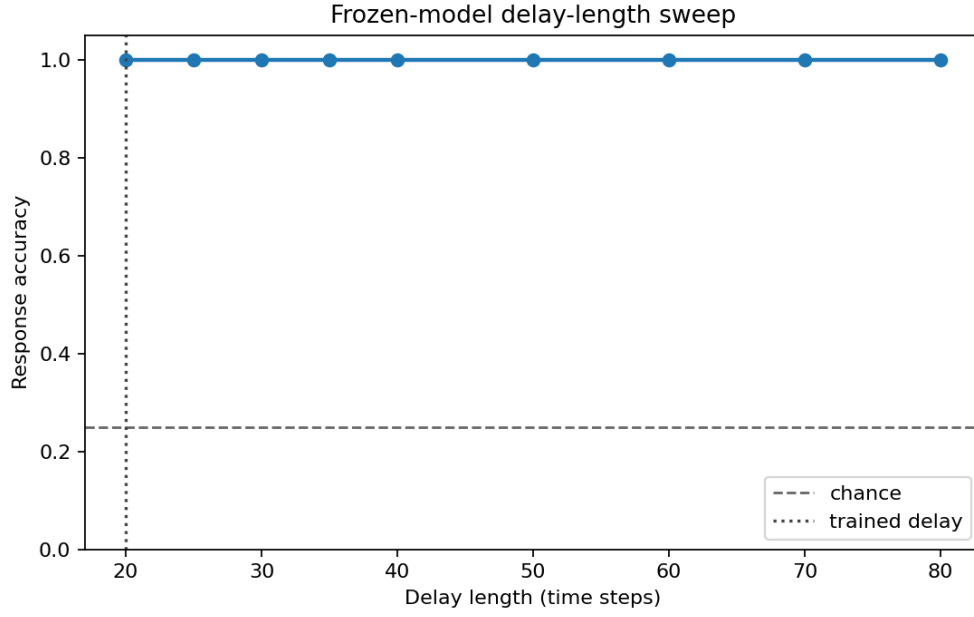


Figure 1: Baseline-delay delay sweep. The tanh categorical baseline remains at perfect accuracy across the tested delays.

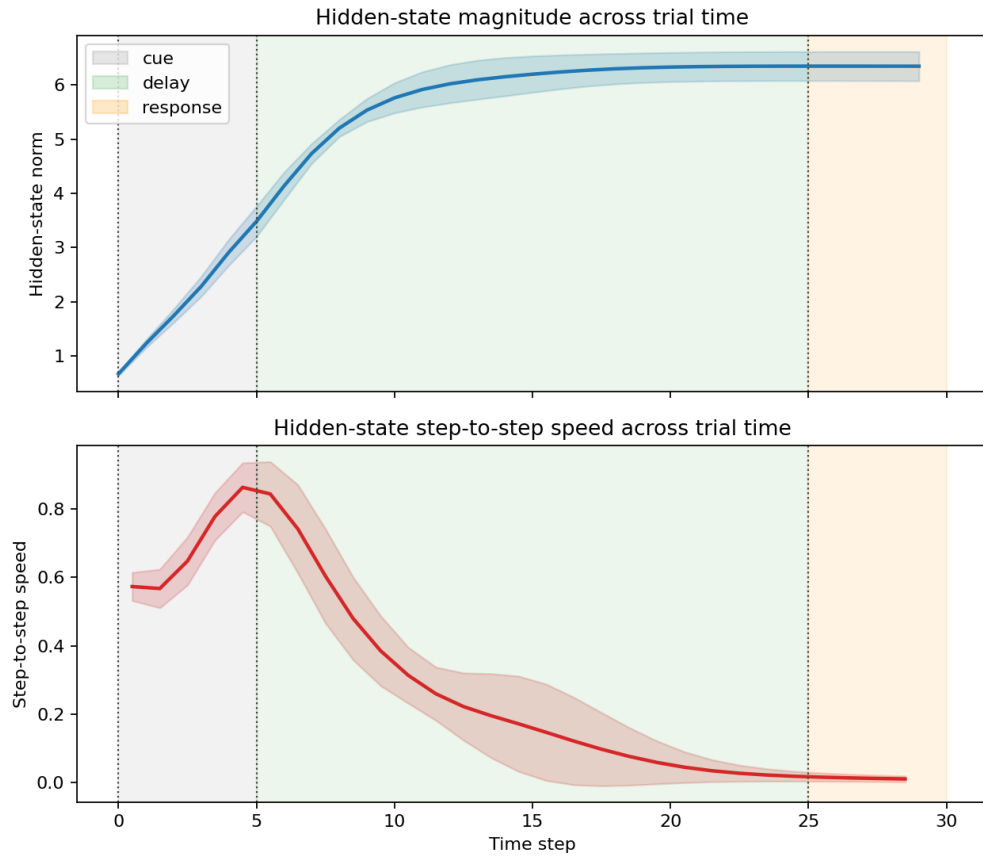


Figure 2: Baseline-delay hidden-state stability analysis. Hidden-state speed decreases during the delay, consistent with settled tanh recurrent dynamics.

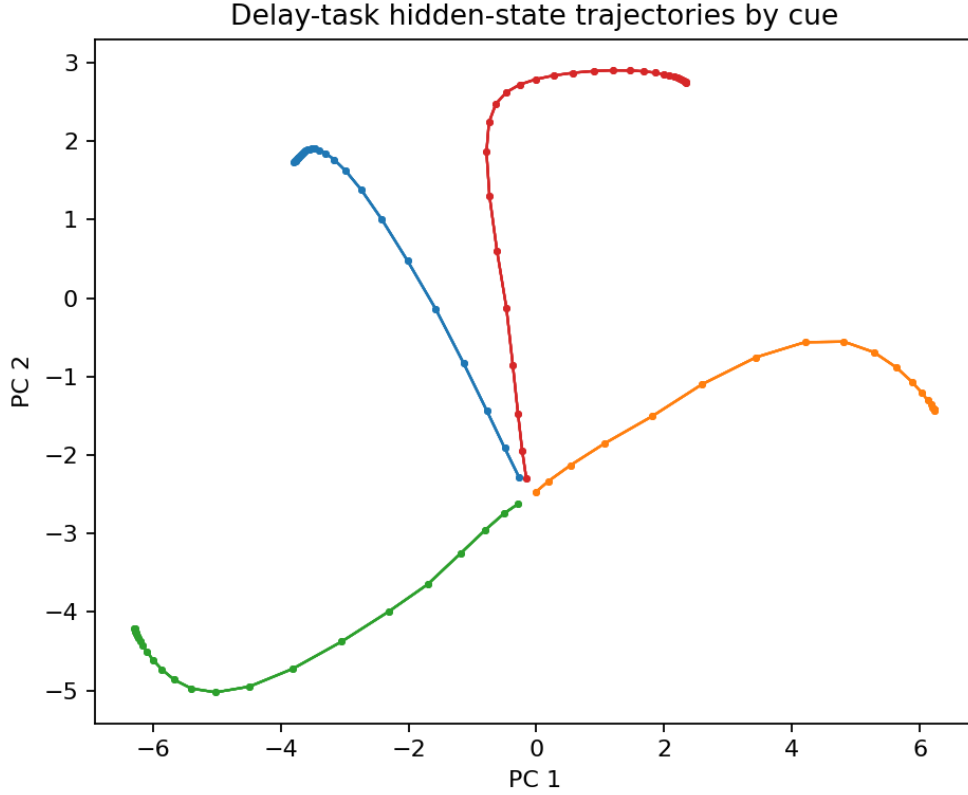


Figure 3: Baseline-delay PCA trajectories. This gives a low-dimensional view of the categorical model’s task-evoked hidden-state trajectories.

3 Tuned Delay Stable

The tuned-delay-stable model was introduced to represent a remembered continuous circular location using tuned units. Each unit has a preferred angle, and cue/target activity is a population bump over those preferred angles. This model preserves angular memory with low error and shows stronger attractor-like evidence than the categorical baseline.

Metric	Value
Held-out mean angular error	0.644 degrees
Held-out median angular error	0.641 degrees
Population MSE	7.08×10^{-5}
Mean error at delay 20	0.643 degrees
Mean error at delay 80	1.453 degrees
Mean error at delay 120	1.903 degrees
Early delay speed	0.0659
Late delay speed	0.00270
Delay settling ratio	0.0409

Table 3: Tuned-delay-stable behavioral and stability metrics. The low delay settling ratio indicates that hidden-state movement slows strongly during the delay.

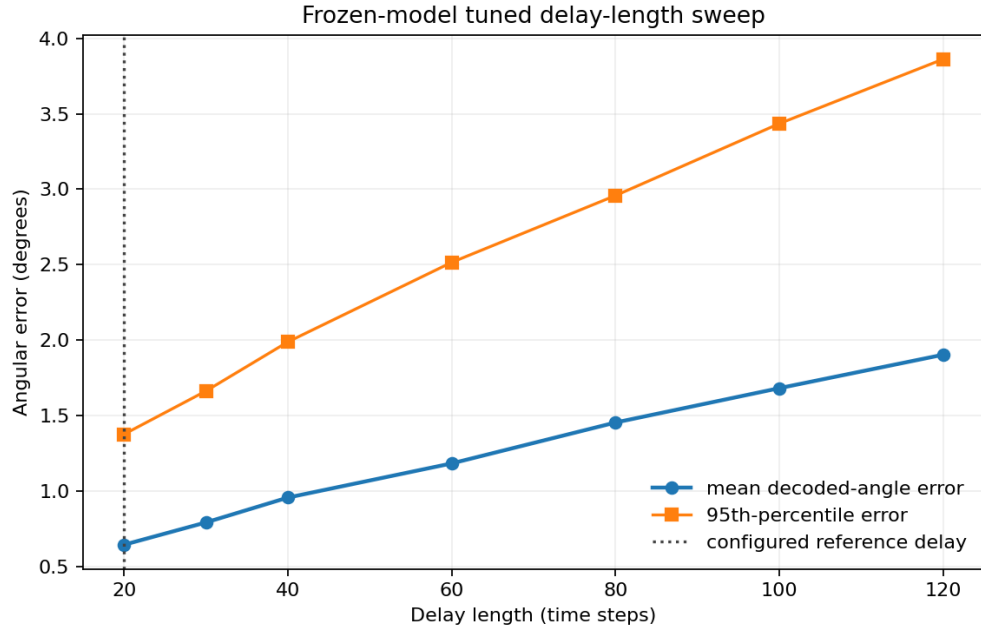


Figure 4: Tuned-delay-stable delay sweep. Mean angular error remains low and rises gradually as the delay extends beyond the trained range.

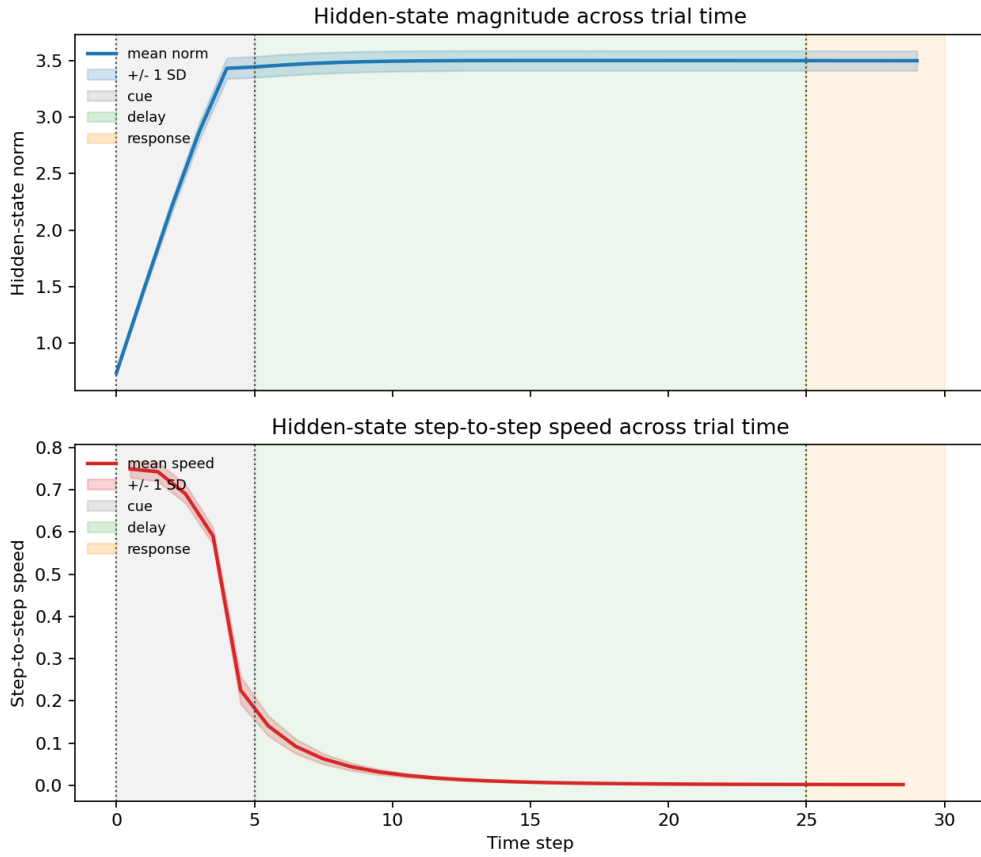


Figure 5: Tuned-delay-stable stability analysis. Hidden-state speed drops sharply across the delay, consistent with settling onto a stable or slow memory manifold.

4 Attractor Evidence in the Tuned Model

The strongest evidence for an attractor-like representation comes from combining several analyses rather than relying on a single figure. The autonomous probe shows low drift under blank-delay dynamics. Fixed-point optimization finds low-speed states that preserve decoded angle. The Jacobian analysis shows a leading near-neutral mode, which is expected for a continuous ring-like memory where movement along the ring changes the represented angle while most other directions are locally stable.

Metric	Value
Attractor-probe final mean speed	0.00145
Attractor-probe mean autonomous drift	1.509 degrees
Attractor-probe p95 autonomous drift	2.918 degrees
Fixed-point mean residual	0.000792
Fixed-point converged fraction	0.766
Mean fixed-point angle error	1.356 degrees
Mean spectral radius	1.00004
Mean second-largest eigenvalue magnitude	0.984
Tangent alignment with leading eigenvector	0.945

Table 4: Tuned-delay-stable attractor-analysis metrics. These support a sampled ring-attractor interpretation, with the caveat that the full hidden-state space is not exhaustively mapped.

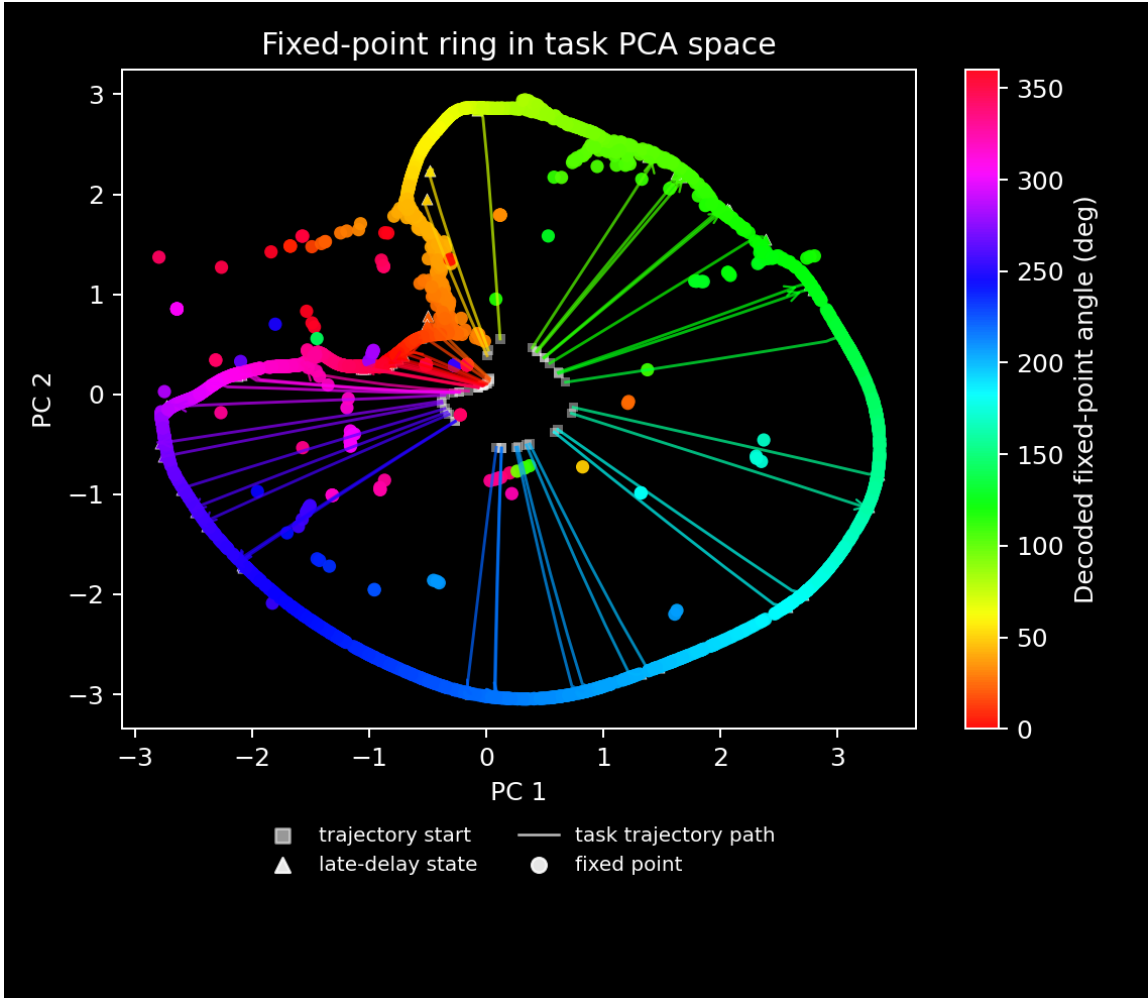


Figure 6: Fixed-point ring in PCA space. Dots are sampled fixed points colored by decoded angle. The task trajectories approach and occupy a ring-like memory manifold. The visible bend or dip is a two-dimensional PCA projection of a higher-dimensional hidden-state structure, not necessarily a collapse of the full attractor.

Converged below 0.001: 0.73

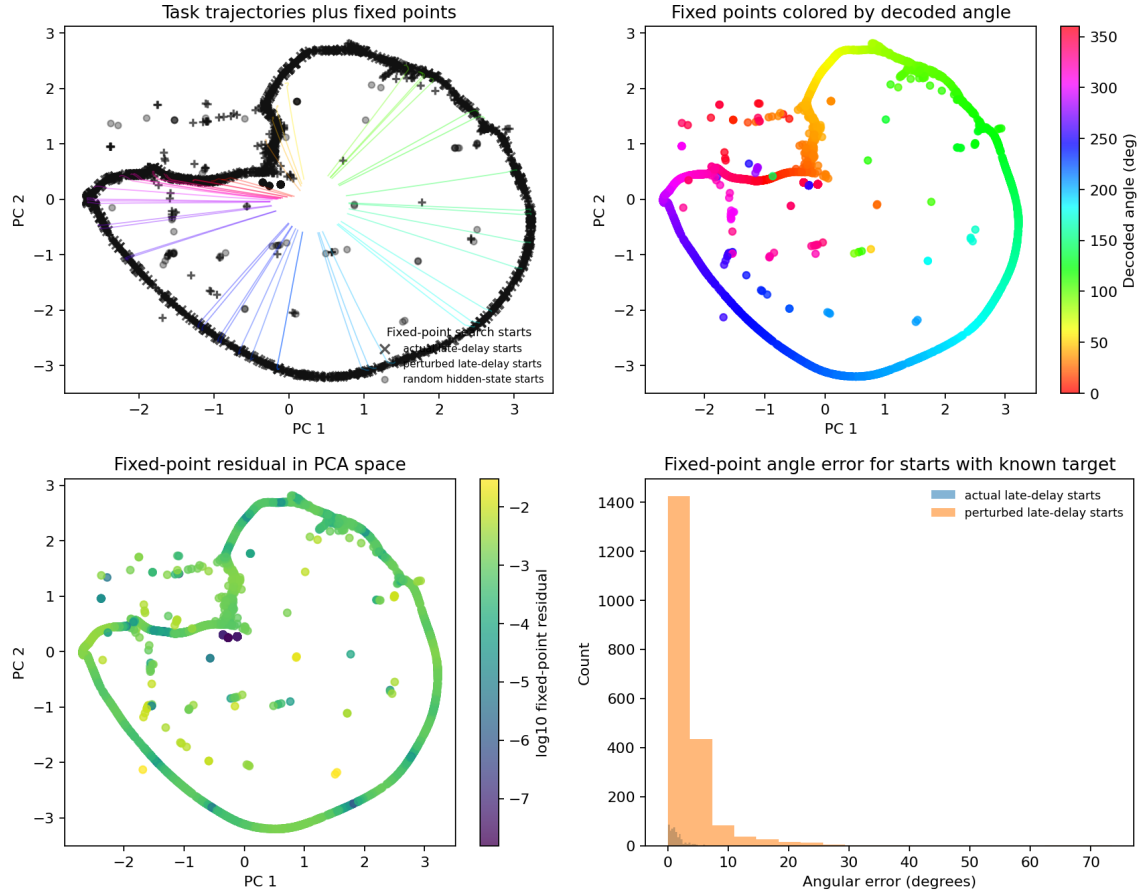


Figure 7: Fixed-point landscape. Top left: task trajectories and fixed-point search endpoints in the same PCA space; black trajectory paths show where normal trials move, while endpoint markers show where searches from actual late-delay states, perturbed late-delay states, and random hidden-state starts converge. Top right: the same fixed-point endpoints colored by decoded angle, showing that position around the ring corresponds to remembered angle. Bottom left: fixed-point residual in PCA space; lower residual means the endpoint changes less under one blank recurrent update, so it is closer to a true fixed point. Bottom right: angular-error histogram for starts with a known target angle; this is mainly a quality-control panel showing how accurately the found fixed points preserve the original remembered angle, not a separate geometry plot. Overall, the panels show that many different start types converge onto a shared ring-like fixed-point structure.

Mean tangent alignment: 0.945

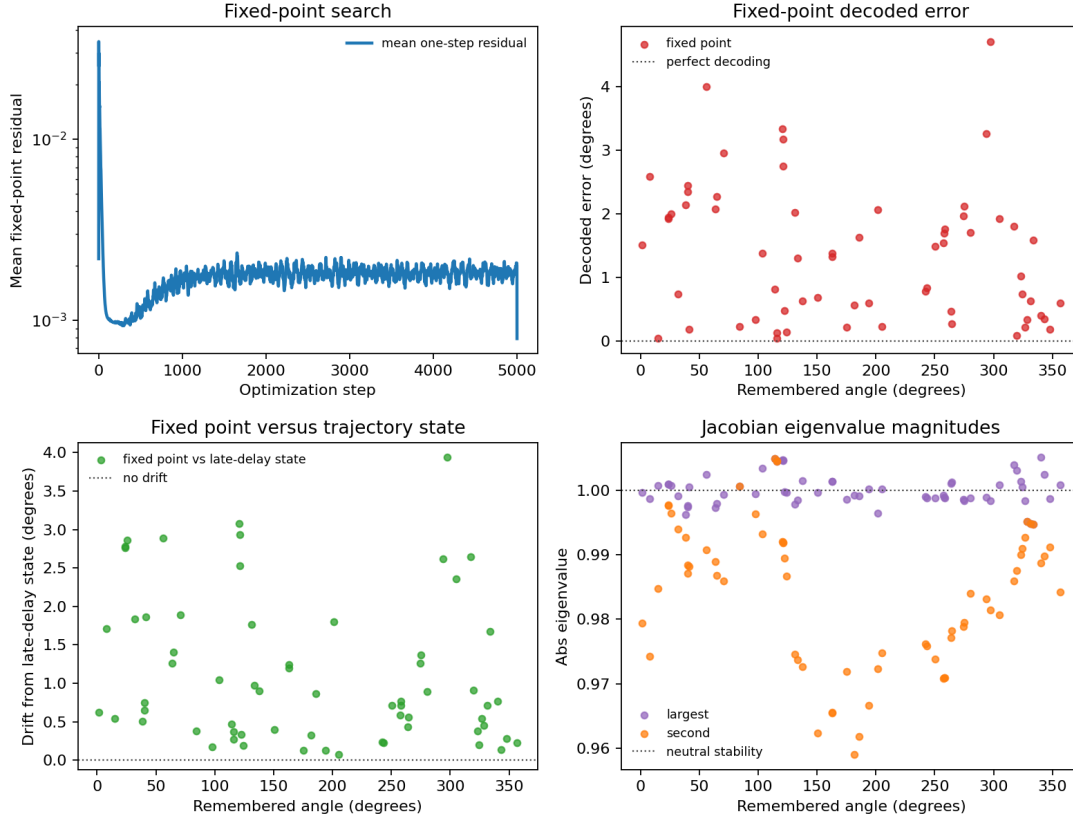


Figure 8: Fixed-point and Jacobian analysis. Top left: fixed-point search loss, shown as the mean one-step recurrent residual during optimization; lower values mean the optimized hidden states change less under blank-delay dynamics. Top right: decoded fixed-point error for each remembered angle; points near the dotted zero-error line mean the optimized fixed point still represents the original target angle. Bottom left: drift between each optimized fixed point and its late-delay trajectory state; small values mean the fixed point lies close to the state the task naturally reached. Bottom right: local Jacobian eigenvalue magnitudes around the fixed points; purple points show the largest local mode, orange points show the second-largest mode, and the dotted line marks neutral stability at magnitude 1. A ring-attractor-like memory should have one near-neutral direction along the ring and more stable directions away from it.

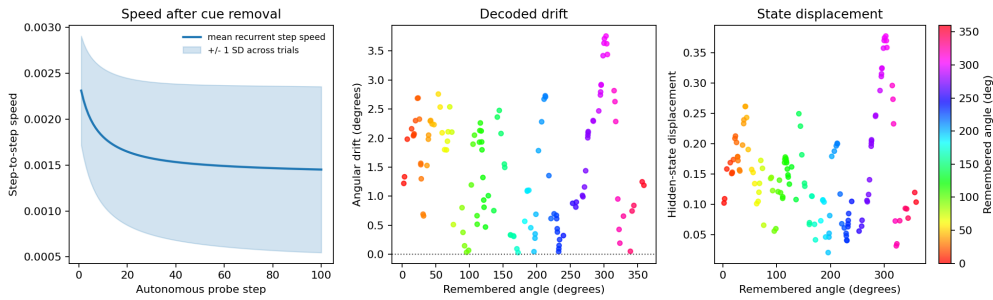


Figure 9: Autonomous attractor probe. Late-delay states continue under blank-delay input. Low final speed and low angular drift support the attractor-like interpretation.

5 Dynamic Visualizations

The movie output is intended as an explanatory aid rather than a new statistical test. It shows the task trajectories moving toward the sampled ring and, in a separate panel, a wide cloud of perturbed late-delay states returning under blank recurrent dynamics. Smooth movie frames are interpolated between real recurrent model steps, so apparent visual smoothness should not be mistaken for additional

recurrent updates.

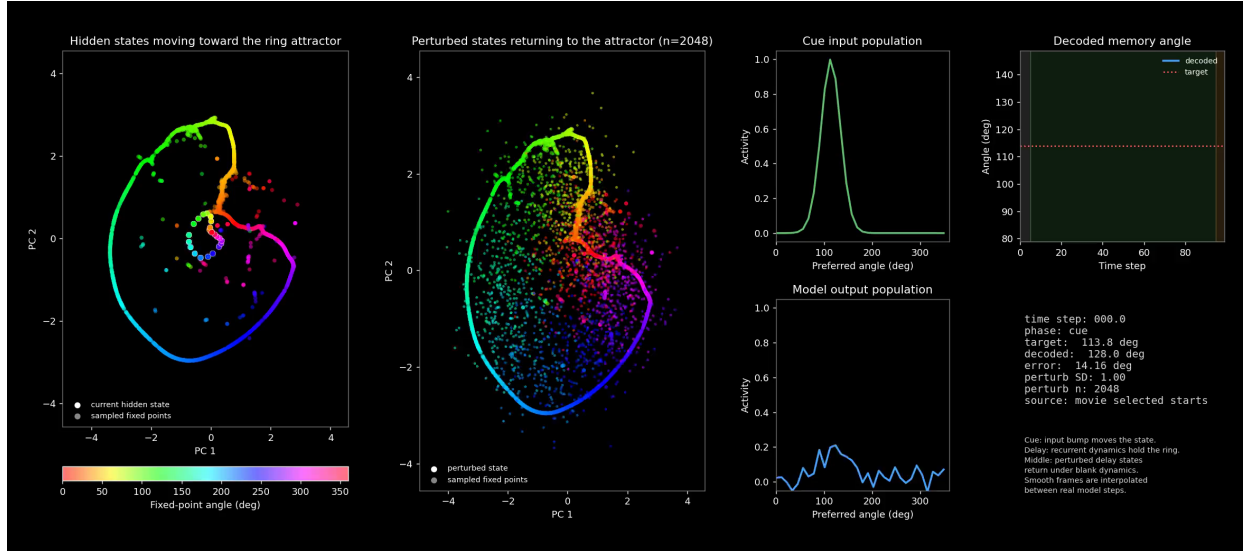


Figure 10: Preview frame from the hidden-state movie. Left: normal task trajectories. Middle: 2048 wide perturbed states returning toward the ring. Right: cue input, model output, decoded angle, and status text. Movie path: `outputs/tuned_delay_stable/figures/tuned_delay_stable_hidden_state_movie.mp4`.

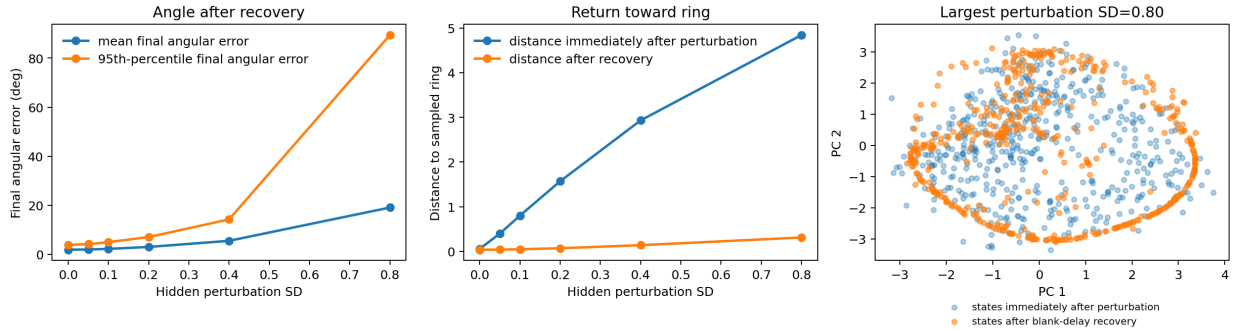


Figure 11: Perturbation recovery summary. Moderate perturbations return close to the sampled ring; very large perturbations produce larger angular errors, showing a finite basin rather than unlimited stability.

6 Comparison and Next Step

The baseline-delay model is behaviorally strong on the categorical task and shows settled tanh delay dynamics. The tuned-delay-stable model is also behaviorally strong, but it adds a continuous circular memory variable and stronger ring-attractor evidence. This makes tuned-delay-stable the better current candidate for the next phase of continuous working-memory modelling.

The next planned development phase is noise and variability analysis. That should be introduced as a separate perturbation experiment, with careful separation between generic stochastic robustness tests and any later psilocybin-inspired modelling assumptions.